

A Cointegration Strategy for the TTF–JKM Spread Using LNG Flow Imbalances

Waterborne Natural Gas trading has become a forefront in the global energy markets over the last 7 years. Traders across the globe move Liquefied Natural Gas (LNG) from continent to continent, both to provide energy for regional demand and to exploit arbitrage opportunities that occur from price differentials. Europe and Asia are the two largest demand destinations for LNG, with the Dutch Title Transfer Facility (TTF) and the Japan-Korea Marker (JKM) being the benchmark futures contract prices for the European and Asian natural gas markets. As LNG cargo has continuously flowed in and out of Europe and Asia over the last 7 years, the two markets have become highly interconnected. Under the efficient market hypothesis, the TTF-JKM spread must remain highly stationary, otherwise rational traders would exploit the arbitrage opportunities that occur and move the spread back to its equilibrium. However, regional natural gas prices react to several stimulus apart from trader activity such as weather shocks, transportation constraints, and geopolitical shocks which all cause existential effects on the TTF JKM spread. Therefore, arbitrage windows persistently exist for natural gas traders. This paper serves to research the dynamics of these spread deviations. I will test whether TTF and JKM are cointegrated price time-series, and search for a reliable signal that can forecast spread deviations from equilibrium. My hypothesis is that if TTF and JKM are cointegrated, then imbalances in LNG inflows between Europe and Asia provide a directional signal for mean reversion trades on the TTF-JKM spread. If LNG flows disproportionately to either continent, the explicit imbalances in supply and demand reflect real-time arbitrage windows that drive an efficient global market back to long-run equilibrium.

To test this hypothesis, I obtained TTF and JKM time-series from Investing.com, along with European and Asian LNG import data from the KPLER Commodity Terminal. The data contains observations from 01/01/2019 to 03/29/2026, giving me 1819 days to perform computations on. I uploaded this data to a Python Jupyter Lab Notebook and compiled the data into a Pandas DataFrame. First, I converted each price time series to log prices to better reflect percent changes and calculated daily spread z-scores. Before I could implement EU and Asian LNG imports, I first needed to test TTF and JKM for cointegration. I first ran two Augmented Dickey Fuller (ADF) tests on each log price series. One tested the null hypothesis that raw log prices are stationary, and the other tested the null hypothesis that daily log price first differences are non-stationary. For the TTF futures contract, the first ADF test returned a p-value of 0.6139, failing to reject the null hypothesis, and the second returned a p-value of 0.0 rejecting the null hypothesis. Similarly, for the JKM futures contract, the first ADF test returned a p-value of 0.6577 failing to reject the null hypothesis, and the second returned a p-value of 0.0 rejecting the null hypothesis. This confirms that the two-price series are integrated $I(1)$ and eligible for cointegration. Next, I performed an Engle-Granger cointegration test, which returned a test statistic of -7.2619 and a p-value of 0. This result strongly rejects the null hypothesis of no cointegration relationship. This analysis provides confirmatory evidence for the first part of my research hypothesis, that the TTF-JKM spread has a strong long-run mean reversion relationship.

Next, I conducted statistical analysis to test if LNG inflow imbalances between Asia and Europe provide a reliable signal for mean reversion trades on the TTF-JKM spread. To begin, I

performed an OLS regression with TTF log prices as the dependent variable and JKM log-prices as the independent variable. The regression returned a $\beta = 1.139$ for $\log_JKM$. In a mean reversion trade, this value represents the hedge ratio to be used when determining position sizing. For every dollar allocated to JKM, a position of 1.139 dollars in TTF is required to proportionately capture the spread mean reversion. The regression also returned 1567 residual values denoted by

$$resid_t = \log(TTF) - \beta \log(JKM) ,$$

which I placed in my DataFrame. The residual represents spread deviation from its long term equilibrium. When determining if a factor's mean-reversion signal is strong, one can test whether that factor's tail values result in a high probability of residual tail values. As a result, I computed the daily z-scores for each residual value. This paper focuses on net LNG inflows, which I defined as $AsiaLNG\ imports - EuropeanLNG\ imports$. The logic behind this signal is as follows: 1) LNG inflows to Asia are substantially greater than those in Europe. 2) This represents an increase in demand for JKM relative to TTF. 3) The TTF-JKM spread tightens. 4) Execute a long position in JKM and a short position in TTF to capture mean reversion in the spread according to the results of the cointegration test. To test this, I first calculated 10 day moving averages for Asian and European LNG imports. Then I created a variable titled "net_flow" defined as $Net_{flow} = Asia10ma - EU10ma$. Subsequently, I calculated z-scores for Net_{flow} . To test Net_{flow} as a predictive signal for residuals 5 days from time = t, I ran an OLS Regression given by the equation

$$resid_{t+5} = \beta * resid_t + Y * net_flow_z_{t-3}.$$

As a result, net_flow_{t-3} had a $Y = -0.0135$, a t-statistic = -4.378, and p-value < 0.001, providing strong evidence that net_flow_{t-3} has highly statistically significant influence on predicting a residual 5 days later. The $Y = -0.0135$ shows when LNG flow to Asia outpaces LNG flow to Europe, the TTF-JKM spread moves downward. If $net_flow_z_{t-3}$ increases by 1, the residual will decrease by 0.0135 5 days later. This value represents arbitrage forming in real time. An important observation to address is that $resid_t$ had a t-statistic of 41.18 and a p-value < 0.001. This intuitively shows that one of the best predictors of the spread tomorrow is the spread today, and as a result, the regression contains extremely high autocorrelation. This makes my model accurate at predicting future residuals when they drift around 0. However, the most interesting occurrences are when the spread makes extremely large jumps or sign changes. This is where the true alpha exists for traders, and may require a model that contains additional factors such as regional temperature anomalies, gas storage levels, and LNG shipping rates. In figure 1 on page 4, I plotted the actual t+5 residual value along with the predicted t+5 residual value from the regression above. It is explicit that LNG flow dynamics are a reliable signal of spread arbitrage opportunity, but also that the model falls slightly behind during high residual volatility regimes.

After determining that the two futures contracts are cointegrated, and LNG flow imbalances are a reliable signal, the strategy to exploit a short spread dislocation regime is as follows: 1) The regime is activated when $spread_{z_t} > 1$ (significant mispricing of the spread) and $net_{flow}_{t-3} > 0$. This means that TTF is overpriced relative to JKM, and LNG is flowing towards Asia. 2) The spread is expected to tighten, so take a long position in JKM and a short position in TTF. These positions must

be sized according to the estimated hedge ratio from my regression. 3) In my regression, the predictive period was for the residual 5 days later, so hold the position for 5 days and exit. For a long spread dislocation regime, the logic is reversed, but the holding period remains the same. To test the effectiveness of this strategy, I implemented a Python script that backtested this strategy over my DataFrame's time interval. The test returned a win rate = 64.29%, Average PnL per trade = 0.039952, Median PnL per trade = 0.01867, and total PnL = 2.094. In figure 2, I plotted the equity curve for this backtest over the time interval. The next question is how to optimally size these positions. A common method is the Kelly Criterion, a formula that calculates the optimal percentage of capital to risk given your probability of winning. The formula is given by

$$f = (p * b - q) / b$$

where f is the fraction of total capital to risk, p is the probability of winning, b is the decimal odds – 1, and q is the probability of losing. Given the backtest's win rate of 64.29%, the resulting optimal size is 28.48% of total available capital. It is well known that Kelly is highly aggressive, and traders often size at $\frac{1}{2}$ or $\frac{1}{4}$ Kelly. For this strategy, I select a size of 7.12% ($\frac{1}{4}$ Kelly) of total available capital to allocate to this trade. As mentioned, this capital must be allocated proportionately to my estimated hedge ratio $\beta = 1.139$. This is found solving the equation $7.12\% = w_{JKM} + 1.139 * w_{TTF}$ resulting in a size of 3.33% for the JKM position and 3.79% for the TTF position.

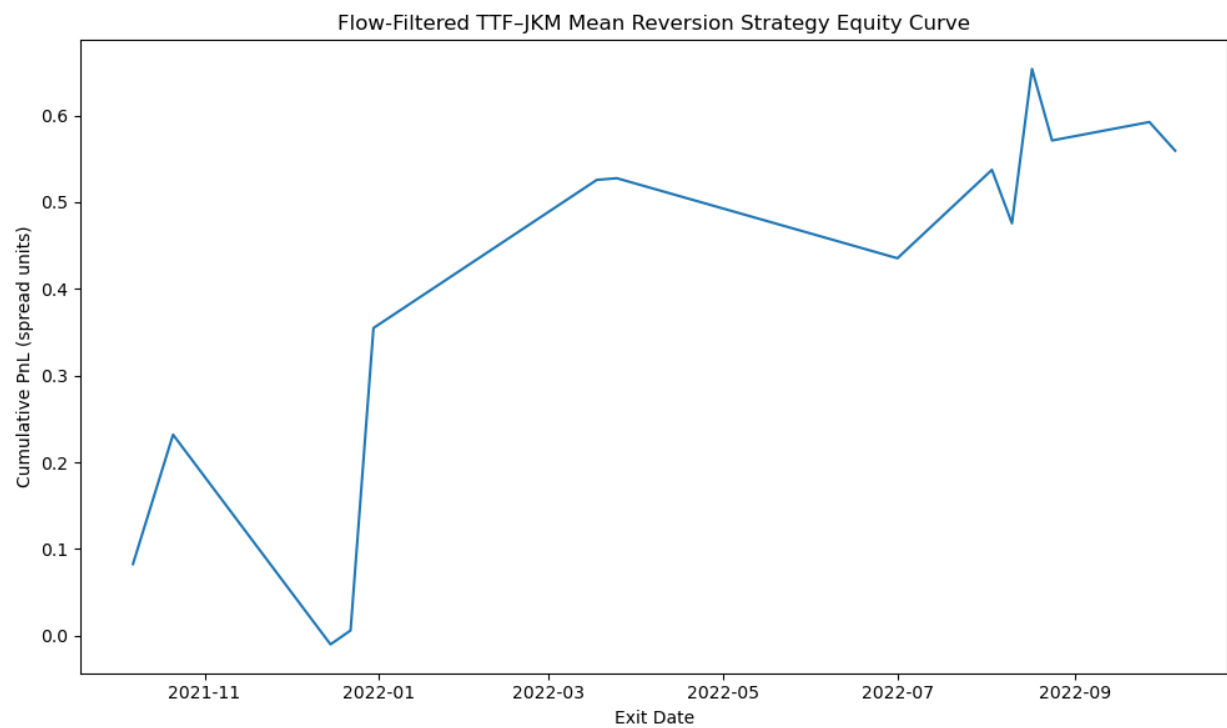
While the results of this research provide strong evidence that LNG flow imbalances are a reliable signal for TTF-JKM spread deviations, several limitations need to be acknowledged. First, the analysis relies purely on LNG import flows, when in reality, LNG markets are influenced by additional factors such as regional storage levels, weather volatility, and geopolitical shocks. Second, my strategy depends on a fixed 5-day holding period however, spread reversion periods are dynamic and are affected by volatility regimes and the magnitude of initial spread dislocation. The model accurately predicts spread movement given normal conditions, however, it struggles to predict extremely large changes in spread, where true alpha exists. With more time, incorporating weather data into my model could potentially improve predicting those extreme events. Lastly, the backtest was conducted purely on my historical price data and ignores transaction costs, bid-ask spreads, and constraints on executing trades.

Despite the limitations above, the findings of this paper provide insight into the physics of global LNG markets. My analysis confirms that TTF and JKM prices are cointegrated, and their spread possesses a stable long-run equilibrium relationship. Most importantly, my analysis shows that LNG flow imbalances between Asia and Europe provide a reliable signal for predicting TTF-JKM spread deviations, capturing the physical mechanism that forms arbitrage opportunities in real time. This paper shows that combining statistical cointegration with physical LNG flow data produces a thorough framework for identifying and exploiting arbitrage windows in global natural gas markets. The findings of this paper ultimately demonstrate that successful trading strategies in the global LNG market require both financial (cointegrated prices) and physical (LNG flow imbalances) signals to reliably achieve profitability in the long run.

Figure 1



Figure 2



Python Notebook, Historical Price Data, EU and Asian LNG import data

https://drive.google.com/drive/folders/1pjbTgVTUWxWra-l2VzumzhiHUf6Tw-3C?usp=drive_link

Note: To run this notebook, you must download and insert all files into your notebook before running.